

Tamogh Nekkanti

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Professional Summary

Machine Learning Engineer specialising in large-scale ML infrastructure, inference systems, and distributed platforms. Experienced building production ML platforms, Kubernetes-native deployment pipelines, LLM serving infrastructure (KServe, NVIDIA Dynamo, vLLM, TensorRT-LLM), and high-performance backend services in Rust and Python.

Work Experience

Machine Learning Engineer at BlackRock

Jul 2025 - Present

Gurgaon

- Engineered a unified machine learning platform supporting end-to-end training, tuning, deployment, and lifecycle management across multi-cloud environments, improving model deployment efficiency.
- Architected and deployed model serving workflows using KServe and NVIDIA Dynamo, enabling multiple runtimes including vLLM, Triton Inference Server, TorchServe, and TensorRT-LLM, increasing runtime flexibility.
- Developed high-performance Rust-based inference orchestration and job scheduling services using Domain-Driven Design (DDD), facilitating automated deployment generation and submission to NVIDIA Dynamo controllers.
- Created deployment framework automating packaging of agent code, model assets, dependencies, and serving runtimes into reproducible container images, accelerating production inference rollout.
- Built production-scale inference infrastructure with KServe and KEDA, enabling event-driven autoscaling that optimized resource utilization.
- Launched inference-as-a-service platform integrating deployment, monitoring, and observability through Grafana and Prometheus, enhancing system reliability and performance tracking.
- Engineered distributed container image build pipeline using Argo Workflows with controller-worker architecture, webhook event propagation, Azure Container Registry integration, and Wiz security scanning, reducing build times.
- Implemented resumable upload service with tusd, supporting large file transfers with pause/resume and extended persistence, improving data transfer robustness.
- Optimized build infrastructure to maintain memory usage under 20% on 500m CPU, enhancing resource efficiency and reducing costs.

Software Developer Intern at BlackRock

Jan 2025 - Jun 2025

Gurgaon

- Developed and optimized ETL pipelines for financial data, forming the backbone of a production-grade trade library, improving data processing speed.
- Revamped legacy systems by integrating large language model (LLM) capabilities into data workflows, enhancing data quality and processing automation.
- Built test server framework to accelerate creation and validation of ETL pipelines, reducing manual setup time.

Open Source Developer (Google Summer of Code 2024) at INCF

May 2024 - Aug 2024

Remote

- Integrated Bayesian inference methods including Markov Chain Monte Carlo (MCMC) and Variational Inference into Neuroptimus, enabling advanced neuronal parameter estimation and improving accuracy.

- Implemented global optimization techniques and custom loss functions to enhance convergence speed and model accuracy.

Student Engineer Intern at ThaparSat – ISRO Program

May 2022 - May 2024

Patiala, India

- Developed soil moisture prediction model leveraging convolutional neural networks (CNN) and satellite-derived features, achieving high prediction accuracy.
- Engineered denoising algorithms for satellite data and devised payload compression algorithms, improving data quality and transmission efficiency.

Projects

Chemical Reaction Prediction (Transformers + GNNs)

May 2024 - Nov 2024

Designed a hybrid architecture combining Graph Attention Networks and Transformers. Built local and global molecular embeddings to capture structural and relational properties. Achieved 74% prediction accuracy on reaction outcomes.

Technologies: Graph Attention Networks, Transformers, Molecular Embeddings

Bayesian Inference Integration (Neuroptimus) [\(Link\)](#)

May 2024 - Aug 2024

Implemented MCMC, Variational Inference, and probabilistic models for neuronal parameter estimation.

Technologies: MCMC, Variational Inference, Probabilistic Models

Motion Amplification & Frequency Analysis (SIH 2023 Finalist) [\(Link\)](#)

Dec 2023

Implemented phase-based motion amplification and frequency extraction from videos using inverse Fourier transform. Achieved 88% accuracy in signal detection.

Technologies: Phase-based Motion Amplification, Frequency Extraction, Inverse Fourier Transform

Education

B.E. in Computer Science and Engineering

Thapar Institute of Engineering and Technology, Patiala, India

GPA: 8.91 / 10.0

- Merit Scholarship Recipient
- Amateur Radio License. Callsign VU3TKI

Skills

Python, Rust, Scala, C, C++

PyTorch, TensorFlow, Scikit-learn, Transformers, NLP, Computer Vision, Graph Neural Networks, Bayesian ML, MCMC, Variational Inference

KServe, NVIDIA Dynamo, vLLM, Triton Inference Server, TensorRT-LLM, TorchServe

ETL, Apache Spark, Hadoop, Airflow, Luigi

Kubernetes, Microsoft Azure, Docker, Argo Workflows, KEDA, CI/CD, Prometheus, Grafana

Microservices, Domain-Driven Design (DDD), SOLID principles, REST, gRPC

PostgreSQL, MySQL, Minio, S3 blob store